

Technical notes of interest to Marine Engineers

Running & servicing pumps !

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General instructions:

Check the pump immediately after receipt. In case of damages or defects - complain immediately. All pumps should be marked with indication of type and number, stamped into the plate as well as into the pump itself - as for centrifugal pumps, usually in the suction flange - as for other pumps, at an easily visible place. If the pump is not mounted immediately after receipt it should be protected during storage.

Mounting instructions:

When calculating the piping system, special regard should be paid to minimizing pressure losses in the suction line (short pipe lengths, large diameter, few bends, no valves). If valves are necessary, then use valves with min. pressure losses, e.g. slide valves, butterfly valves, flap non-return valves etc. Static suction lift + total pressure losses in suction line + NPSH value of the pump + steam pressure of the liquid must not exceed 10 m WC at sea level (and 760 mm Hg).

For pumps delivered without motor

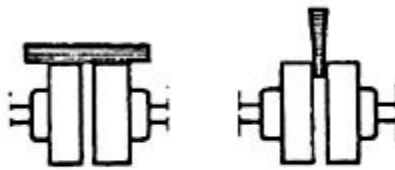
Mounting of coupling parts:

Before mounting, shaft and coupling parts must be carefully cleaned. Grease shaft ends and check that key fits. Heat coupling parts to 70°C and push

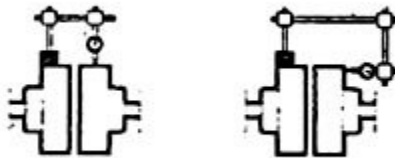
them on to the shaft ends by hand or by a light pressure. By dismounting a puller will have to be applied.

Alignment of couplings:

Flexible couplings will absorb small deviating in the relative positions of the shaft ends to be connected, however, a careful and accurate alignment will prolong the life of the coupling flexibles. When aligning the coupling halves the parallel and angular accuracy should be as great as possible. Alignment must be carried out in two axial planes at right angles (see sketch).



Alignment of coupling parts by means of straight-edge and feeler gauge.



Alignment of coupling parts by means of indicator.

A perfect alignment can be achieved by means of a straight-edge (axial parallelism) and a feeler gauge (squareness) or by means of an indicator, as shown in figures above. The coupling clearance must be within 2-6 mm according to the size of the coupling.

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Installation of the pump:

The entire piping must be pressure and leakage tested and flushed clean before it is connected to the suction and discharge side of the pump. If the pressure test takes place with the pump installed the testing pressure must not exceed the testing pressure of the pump, which is stamped into the pump, usually in the suction flange. Otherwise, the pump's packing or flanges for the mechanical shaft seal might be damaged. Connection of the

pipes to the pump must take place without using force as this might interfere with the alignment of the pump.

Precautions must be taken to ensure that the pipe connections to the pump are supported in a way that prevents tensions from being transmitted to the pump, e.g. owing to thermal expansion. Check that all other connections to the pump are in working order, such as lubrication, priming, heating, cooling medium outlet, etc., and that the various measuring equipment has been correctly connected. Take care that no parts obstruct access to the pump for inspection or replacement of important parts. For pumps with axially split casing there should be sufficient space for dismantling the top part of the pump casing.



Motor wiring:

Electric motors should be connected by means of sufficiently long cables to permit the motor to be moved a little without dismantling the connections. A control unit (emergency stop) should be installed close to the pump.

Engine driven pumps (diesel):

When using pumps driven by internal combustion engines vibrations may occur in the system. It is recommendable to mount the unit on vibration dampers and to install compensators between pump and pipelines. The calculation of vibration dampers should be left to specialized companies.

Starting of the pump:

When the preceding preparations and installations have been completed the pump can be started. Centrifugal pumps must be provided with a bypass if it is operated for long periods with closed valve on the discharge side as, otherwise, a too strong heating and expansion of the pump medium will occur. When starting positive pumps (e.g. piston pumps, gear pumps and others), the valve on the discharge side must be open as, otherwise, the pump may burst, in spite of the fact that a bypass valve to the suction side is incorporated in these pumps.

Pumps provided with mechanical priming pump should be checked for supply / contents of liquid before starting. - for further information see the OEM's operation instructions for priming pumps.

Pumps with mechanical shaft seal must be protected against impurities and dry running. Supply of liquid or oil to the mechanical shaft seal is imperative. Please note that clogged filters will cause considerable friction losses, which may decisively affect the suction capacity and output of the pump.

Check also that the pump shaft can be turned by hand. Check also when starting the pump that the direction of rotation is correct (usually shown with an arrow on the pump), and open slowly the valve on the discharge side of the pump. At normal discharge head the pump should run smoothly without abnormal noises and vibrations after about one minute of operation.

Pumps with priming device:

The valve on the discharge side of the pump must be closed during priming. The operating instructions for the priming system should be strictly followed, see instructions for the system in question (mechanical system with pump or ejector system driven by compressed air). After priming of the pump, open the valve on the discharge side slowly and adjust it in such a way that permissible vacuum is not exceeded.

Self-priming pumps:



All pumps:

It is an absolute requirement that pumps are not operating unnecessarily long without liquid, normally max. 5 min. During the first hours of operation bearings, packings and mechanical shaft (seal(s)) should be checked for heating and leakage. Normal bearing temperature is 40-75°C. Max. temperature for normal ball and roller bearings is up to 105°C.

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Maintenance of pumps:

Lubrication:

To pumps directly coupled on motor shaft it applies that only the bearings of the motor require lubrication.

Pumps designed with bearing bracket and grease lubricated bearings should be lubricated every 3rd month with ball bearing grease (lithium soap grease - MOBILUX EP2 or similar quality). Check that not too much grease is pressed into the bearings as this will cause generation of heat. All bearings are provided with sealing rings in order to prevent humidity and impurities from entering. The grease must not be pressed through these sealing rings. For bearings which are exposed to higher pressures or humidity a heat resisting and water-repelling grease will apply (MOLYPAN LT-1, Petro Canada Precision, or similar quality).

For pumps designed with oil lubrication, lubricating oil (MOBILGEAR 682 or similar quality) can be used for normally loaded bearings. For more heavily loaded bearings DTE26 or similar quality should be used. In case of oil sump lubrication the oil level should be checked by means of screw-plug, oil-level glass, oil indicator or other control equipment. see instructions for the pump in question.

3 months' check:

Check the pump for false noises, vibrations and leakage. Pumps with mechanical shaft seal must not leak, neither during standstill with inlet pressure on the pump, nor during operation. The bypass pipe to the mechanical shaft seal must be intact, and permit free flow. Likewise, the piping system and filter of the priming device must be intact and permit free flow. In case of defects they must be corrected immediately as they may greatly influence the operation and life of the pump.

12 months' check:

Disassemble the pump (see instructions). Check the components of the rotary element and perform all necessary repairs. Normal clearance between impeller and sealing rings is 0,2-0,3 mm, and max. clearance is 0,5-0,8 mm, according to size and capacity. Clean the pump casing inside and paint it if necessary (e.g. with APEXIOR No. 3 or with more corrosion-resistant products, e.g. HEMPADUR No. 1513 "Hempel Marine Paints").

Disassembling instructions:

Before starting the disassembly of the pump the sectional drawings of the various pump parts should be studied closely. Most pumps are designed for easy disassembly, and no special tools should be required. The pumps consist of standard components, which means a large number of the components are used in various pump sizes, for which reason it is necessary to keep a small stock of spares for various sizes of the same pump type.

Pumps with split casing, can be disassembled without dismantling motor and pipe connections. After dismounting of coupling bolts (tripartite coupling), bearing housing top part and pump casing top part, the rotating element consisting of shaft with impellers(s), bearings, casing wear rings, mechanical seal(s) and coupling half can be lifted out for inspection, repair or replacement.

Assembling instructions:

When assembling the pump the mechanical seal must be treated with the greatest care. Take care that the various parts are placed correctly. The rubber bellows is mounted on the shaft by means of grease, which gives approx. two hours respite before the bellows jams on the shaft. This means that within this period it can be displaced when the packing flange is clamped by means of two or four nuts. Otherwise, the mechanical seal might be damaged. Before mounting the O-rings (if any) they must be cleaned and provided with a thin layer of grease. For further information see instructions for mechanical seal.

Bearing packing which was removed, must be reinstalled. In such places where liquid packing has once been used this type of packing must be used again when remounting the pump, as other types of packing would produce wrong spaces.

Mounting of motors on direct-coupled pumps:

If the pump is delivered without motor a distance ring is mounted between the packing flange and the shaft. This ring must not be removed until the motor has been mounted and the pump shaft screwed together with the motor shaft. Do not forget to remove the ring before starting the pump!

Pumps with split casing:



When mounting the rotating element in the pump, take care that the tongue of the wear rings fits correctly in the groove of the pump casing. The bolts of the pump casing must be tightened with a factor of 6,0 kg/m, the bolts of the bearing housings with a factor of 3,0 kg/m.

When reassembling pumps, take care that impellers, as regards the vanes, have the same direction of rotation as before the disassembling.

Maintenance (Storage instructions):

General instructions:

All pumps which are taken out of operation for some time should be protected against corrosion and frost by being kept in a dry place.

Additionally, the following directions should be observed:

Preservation of bearings:

Both ball / roller bearings and slide bearings should be lubricated all over with their normal lubricant.

Preservation of interior parts of the pumps: The interior parts of the pumps should be protected against aggressive pumping media, for instance by using a water-repelling solvent containing at the same time corrosion inhibiting nitrides. We recommend MOBIL OIL - ARMA 245 or similar quality.

Preservation of gear wheel pumps and gearboxes: Gear wheel pumps do not need any protection if they have been operating with oil products. If not, they should be lubricated all over with a neutral oil. Gearboxes should be completely filled with the oil specified. If the standstill lasts longer than 12 months the oil should be emptied and replaced by MOBILARMA25 or similar quality and the gearbox should be flushed with this oil.

Preservation procedure:

Flush the pump with pure, hot water (does not apply to screw pumps and gear pumps) and let it airdry. Close the valves on both sides of the pump and flush the pump with MOBIL ARMA 25 or similar quality. Flushing can take place through the manometer connections.

Restarting of the pumps:

When restarting pumps it is advisable to fill the pump with hot water through the manometer connections (does not apply to screw pumps and gear wheel pumps) and to turn the pump shaft by hand before starting the pump. Watch the operation of the pump for about 10 minutes and open the valve in the normal way. Bearings, gearboxes etc. should be lubricated with normal oil or grease if a change of oil has taken place.

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Trouble shooting

Before starting any particular troubleshooting, you should check these basic items.

- Check direction of rotation (marked with an arrow).
- Check that there is liquid in the pump (might have been emptied by standstill).
- Check that the pump is able to turn by hand.
- Check that the pump has been lubricated with grease.
- Check the pump for noises and vibrations immediately after starting.
- Check priming pump (if any). See special instructions.

Possible problems...

- A. [Pump and motor cannot be actuated](#)
- B. [Motor running but no pumping effect](#)
- C. [Insufficient capacity](#)
- D. [Pump pressure too high](#)
- E. [Capacity too large](#)
- F. [Discharge head too low](#)
- G. [Power consumption too large](#)
- H. [Pump output decreases or stops](#)
- I. [Irregular running](#)
- K. [Increasing noise level](#)
- L. [Leaks](#)
- M. [Bearing temperature too high](#)
- N. [Pump wears out quickly](#)
- O. [Stark vibrations](#)

A. Pump and motor cannot be actuated:

1. Impeller or shaft blocked.
2. Motor fault.

B. Motor running but no pumping effect:

1. Motor rotation is not transmitted through coupling.
2. Discharge valve closed.
3. Non-return valve or other valves are closed.
4. Suction line closed or filter clogged.
5. Air in pump casing.
6. Suction line leaking.
7. Shaft seal leaking.
8. Bottom valve defective.
9. Suction lift too high.
10. Priming pump defective.

C. Insufficient capacity:

1. Wrong direction of rotation.
2. Number of revolutions too low.
3. Counter-pressure too high.
4. Suction line or impeller partly clogged.
5. Air in pump casing.
6. Air in pumping medium.
7. Suction lift too high (inlet pressure too low).
8. Capitation.
9. Suction line leaking.
10. Shaft seal leaking.
11. Pump worn out.

D. Pump pressure too high:

1. Number of revolutions too high.
2. Impeller oversized.
3. Too many pressure stages.
4. Specific gravity of pumping medium too high.
5. Viscosity of pumping medium too low.
6. Inlet pressure too high.
7. Manometer defective.

E. Capacity too large:

1. Number of revolutions too high.
2. Impeller diameter too big.
3. Counter-pressure too low.

F. Discharge head too low:

1. Number of revolutions too low.
2. Impeller diameter too small.
3. Too few pressure stages.
4. Specific gravity of pumping medium too low.
5. Viscosity of pumping medium too high.
6. Manometer defective.

G. Power consumption too large:

1. Motor too small.
2. Motor fault.
3. Capacity too large.
4. Counter-pressure too low.
5. Stuffing-box tightened too much.
6. Shaft ends out of alignment.
7. Electricity supply incorrect (voltage, amperage, frequency).

H. Pump output decreases or stops:

1. Suction line leaking.
2. Shaft seal leaking.
3. Increasing suction lift.
4. Filter clogged.
5. Cavitation.

I. Irregular running:

1. Bearings defective.
2. Motor fault.

K. Increasing noise level:

1. Beginning cavitations.
2. Air in pumping medium.

3. Capacity too large.
4. Clamping to base loosened.
5. Base bolts loosened.
6. Influences from pipe connections or base.

L. Leaks:

1. Cracks in pump casing.
2. Faulty assembly of pump.
3. Pipe connections leaking.
4. Shaft seal leaking (in case of soft stuffing-box packing minor leaks are necessary).

M. Bearing temperature too high:

1. Faulty lubrication or wrong lubricant.
2. Deficient pump alignment.
3. Influences from pipe line.
4. Coupling distance wrong.
5. Shaft bent.
6. Foreign bodies or impurities in bearings.

N. Pump wears out quickly:

1. Wrong materials in relation to pumping medium.
2. Cavitations.
3. Stuffing-box tightened too much.
4. Shaft bent.
5. Deficient alignment.
6. Influences through pipe line.

O. Stark vibrations:

1. Foreign bodies in pump.
2. Motor out of balance.
3. Other influences.